COURSE 2 · SECTION 5 · EXERCISE

Stack, safety factor and Cpk

Section 5 (videos S05.1 and S05.2) · Tolerance stacks WC vs RSS, FEA safety factor, Cpk

Start: Section06_1_Start.pbix or your Engxtech_S05_2.pbix; save a copy as Engxtech_S05_Exercise.pbix

Time: about 25 min

Data: the course data folder

DESIGN QUESTION

Does the shaft end float stack (STK-02) meet its requirement by worst case and by RSS? Can the seat ring OD tolerance alone make STK-01 pass worst case? Which load case and which machine have the least margin?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

StackContributors lists each dimension in a stack with Direction (+1 adds to the gap, -1 reduces it), nominal and +/- tolerance. Rev blank = all revs; 'A/B' and 'C' are the seat ring OD rows before and after ECR-3200. The video built the SeatRingBuild selector, the stack measures, the OD Tol what-if parameter, FEA[SF] and the Cpk chart. SF target 1.5; Cpk minimum 1.33.

Your tasks

1. In your exercise copy, on the Stacks page set the SeatRingBuild slicer to A/B and read STK-02: WC Min, WC Max, RSS Min, RSS Max and the two results.
2. Set the slicer to C and read STK-01.
3. Set the OD Tol slider to 0.000 and read WC Max (what-if) for STK-01. Is it below the 0.080 limit?
4. FEA & Cpk page: read Cases Below Target and the lowest SF; read Machines Below 1.33 and the lowest Cpk.
5. New measure in _Measures: **Min Cpk Margin** = Min Cpk - 1.33, 3 decimals. Which machine is closest to the limit without being below it?

Check your answer

Check	Expected
STK-02 worst case (mm)	**-0.020..0.420 (fail)**
STK-02 RSS (mm)	**0.0842..0.3158 (pass)**
STK-01 rev C worst case (mm)	**0.006..0.094 (fail)**
STK-01 rev C RSS (mm)	**0.0298..0.0702 (pass)**
STK-01 WC max with OD tol = 0 (mm, req max 0.080)	**0.086**
Lowest SF	**P200-B Drop 1 m 1.40**
Load cases below SF 1.5	**1**
Machines with Cpk below 1.33	**3**
Lowest Cpk	**GRD-02 0.787**
Closest machine still at or above 1.33	**INJ-02, Cpk 1.412, margin +0.082**

Hint. CNC-M03 has Cpk 1.329. Shown with 2 decimals it reads 1.33, but it is below 1.33: count it, and show Cpk with 3 decimals.

Stuck? The worked solution with the full DAX and M is in Solutions/S05_Solution.pdf.